

Mark schemes

Q1.

(a) (i) $z_1 = \frac{(60 - 40)}{10} = 2$

Method for z (ignore sign, ignore errors in converting 9.00 am into minutes after 8.00 am)

M1

Probability before 9.00 am = 0.97725

a correct use of normal tables

0.977(0.977 – 0.978)

M1/A1

3

(ii) $z_2 = \frac{(30 - 40)}{10} = -1$

Probability between 8.30 am and 9.00 am

$= 0.97725 - (1 - 0.84134) = 0.819$

completely correct method

0.819(0.818 – 0.819)

m1/A1

2

(b) $35 + 2.3263 \times 2 = 39.65$

2.3263(2.32 × 2.33)

their z × 2

B1/M1

Leave at 8.20 am

method for number of minutes before 9.00 am

method for time

8.20(8.20 – 8.21)

m1/m1/A1

5

(c) Maximum if aims for 8.45 arrival time

E2(1)

2

i.e. leave 28 minutes before 8.45 am

8.17 am

(d) (i) Shortest mean time

E1

(ii) Least variable

E1

2

[14]

Q2.

(a) $\bar{x} = 12$ $s = 1.73$

12 cao

1.73(1.73 – 1.735) allow

1.63(1.63 – 1.635)

0.4 cao

B1/B1

2

(b) 0.4

0.4 cao

B1

1

(c) Binomial

B1

1

(d) (i) $P(X > 15) = 0.9029$

attempted use of $B(30, p)$, their p method – allow $P(\geq 15)$

B1/M1

$= 0.0971$

0.0971(0.097 – 0.0972)

A1

3

(ii) Mean 12

12 cao

B1

s.d. = $\sqrt{30 \times 0.4 \times 0.6} = 2.68$

Method – allow variance if called variance, their p 2.68(2.68 – 2.69)

M1/A1

3

(e) Unsuitable

Unsuitable

E1

standard deviations very different

ft s.d. different

max 1 if previous M not earned

E1 ft

2

(Mean same but this would be the case no matter what the distribution of X)

(f) P not constant – some houses more likely to put boxes out than others

P not constant/not independent

E1

1

[13]

Q3.

(a) (i) $z = (185 - 65)/60 = 2.0$
method for z

M1

probability < 185 = $1 - 0.97725 =$
completely correct method

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M1

0.02275

0.0227 – 0.023

A1

3

(ii) $z_1 = (50 - 65)/60 = -0.25$

$z_2 = (125 - 65)/60 = 1.0$

method for both z's including correct sign

M1

Probability between 50 and 125 minutes

any correct use of normal tables

M1

$$= 0.84134 - (1 - 0.59871)$$

completely correct method

M1

$$= 0.440$$

$$0.439 - 0.441$$

A1

4

- (b) Mean only just over one standard deviation above zero.

Normal distribution would give a substantial chance of negative time which is impossible.

allow both marks if point clearly made

B2

2

[9]

Q4.

- (a) (i) $z = (28 - 23)/4 = 1.25$
- attempt at z - ignore sign*
completely correct method

M1

$$P(\text{less than } 28) =$$

M1

$$0.894$$

$$0.894 \sim 0.895$$

A1

3

- (ii) $z_1 = (20 - 23)/4 = -0.75$
- correct method, both z's*

M1

$$z_2 = (30 - 23)/4 = 1.75$$

any correct use of normal tables

M1

$$P(\text{between } 20 \text{ and } 30)$$

completely correct method

M1

$$= 0.95994 - (1 - 0.77337)$$

$$= 0.733$$

$$0.733 \sim 0.734$$

A1

4

(b) $23 + 2.3263 \times 4$

2.3263 or 2.326 or 2.33 or 2.32

any correct expression, their z

completely correct method, their z

B1 M1 M1

$$= 32.3 \text{ minutes}$$

$$32.3 - 32.31$$

A1

4

(c) $23 \pm 1.6449 \times 4 =$

1.6449 or 1.645 or 1.64 or 1.65

B1

$$23 \pm 6.58$$

their z $\times 4$

completely correct method, their z

M1 M1

$$16.4 \sim 29.6$$

6.575 $\sim$ 6.585 ; 16.4 $\sim$ 16.45 ;

29.55 $\sim$ 29.6

A1

4

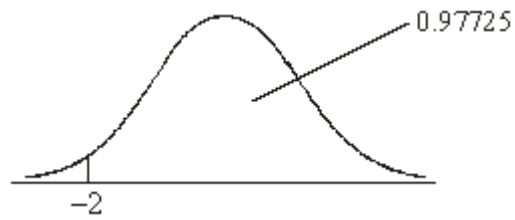
[15]

Q5.

(a) (i) $z = \frac{111 - 135}{12} = -2$

*Attempt at z, ignore sign,
disallow variance*

M1



Any correct use of normal tables

M1

$$P(> 111) = 0.977$$

$$0.977 - 0.978$$

A1

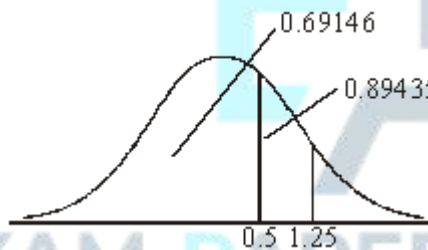
3

(ii) $z_1 = \frac{141 - 135}{12} = 0.5$

$$z_2 = \frac{150 - 135}{12} = 1.25$$

Correct method, both zs, ignore signs

M1



Correct use of normal tables

Completely correct method

m1 m1

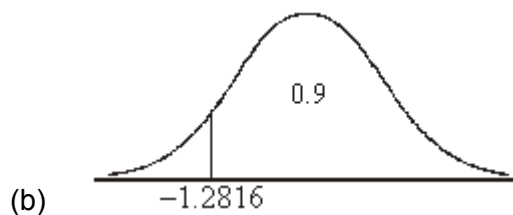
Probability between 141 and 150

$$0.89435 - 0.69146 = 0.203$$

$$0.202 - 0.204$$

A1

4



1.28 – 1.29 ignore sign

B1

Longest journey $135 - 1.2816 \times 12$

Correct method – allow incorrect sign

Completely correct method

M1 m1

= 120 miles

119 – 120

A1

4

[11]

Q6.

(a) (i) Binomial $n = 10$ $p = 0.4$

B1

$P(3 \text{ or fewer}) = 0.382$

Correct use of tables or correct method of calculation

0.382-0.383

M1 A1

3

(ii) $(> 4) = 1 - (4 \text{ or fewer}) = 1 - 0.6331$

$(> 4) = 1 - P(4 \text{ or fewer})$ *or*
correct method of calculation

M1

= 0.367

0.366 – 0.368

A1

2

(b) Mean = $10 \times 0.4 = 4$

cao

B1

Standard deviation = $\sqrt{10 \times 0.4 \times 0.6} =$

Method for s.d. or variance (if called variance)

M1

1.55

$$1.54 - 1.56$$

A1

3

(c) 3.92

$$3.92 - 3.93$$

B1

3.17 (allow 3.05)

$$3.17 - 3.18 \text{ (3.045 - 3.055)}$$

B1

2

(d) Mean similar but standard deviation much greater than model
Standard deviations different

E1

Binomial not good model

Binomial not suitable model

E1

Siballi's beliefs not plausible

Siballi's beliefs not supported

E1

3

[13]

Q7.

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(a) $z_1 = \frac{250 - 266}{32} = -0.5$ Exam Papers Practice. All rights reserved.

Attempt z ignore sign

M1

$$z_2 = \frac{300 - 266}{32} = 1.0625$$



Any correct use of normal tables

Correct method for both z's

M1 m1

Probability between 250 and 300

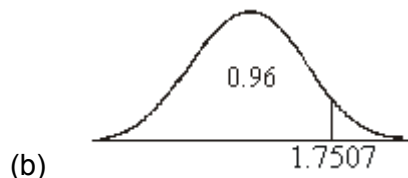
$$0.8560 - (1 - 0.69146) = 0.547$$

Completely correct method

$$0.547 (0.546 - 0.548)$$

m1 A1

5



$$1.7507 (1.75 - 1.751)$$

Method - ignore sign

Completely correct method

B1 M1 m1

$$k = 266 + 1.7507 \times 32 = 322$$

$$322 (321.5 - 322.5)$$

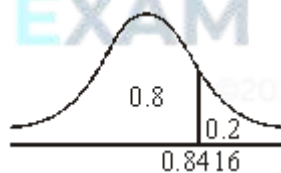
A1

4

[9]

Q8.

(a)



$$0.8416 (0.841 - 0.842)$$

Use of $z \times 6$.

B1 M1 m1 A1

Completely correct method

4

$$25.0 (25.0 - 25.1)$$

(b) (i)

$$\frac{6}{\sqrt{90}} = 0.632$$

$$\frac{6}{\sqrt{90}} \text{ acf}$$

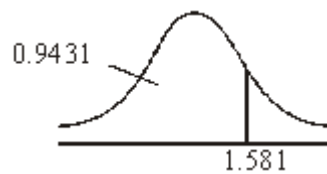
B1

1

(ii)
$$z = \frac{(21 - 20)}{\frac{6}{\sqrt{90}}} = 1.581$$

Use of $\frac{6}{\sqrt{90}}$.
Method for z – ignore sign

M1 m1



Probability mean exceeds 21 is
 $1 - 0.9431 = 0.0569$

Completely correct method
 $0.0569(0.0568 - 0.0571)$

M1 A1

4

[9]